

KARTIK SINGH

+91-9911274999 | justkartik9@gmail.com | [LinkedIn](#) | [Github](#) | [LeetCode](#)

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript | **Machine Learning & AI:** Scikit-learn, TensorFlow, XGBoost, LightGBM, LangChain, Anthropic (Claude) API, Regression, Classification, Feature Engineering | **Data & Databases:** Pandas, NumPy, MySQL, MongoDB, PostgreSQL | **BI & Visualization:** Power BI, Tableau, Streamlit, Excel Dashboards
Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Jenkins, Git | **Backend & Web:** React.js, Node.js, Next.js, FastAPI, REST APIs, GraphQL | **Product & Analytics:** KPI Tracking, A/B Testing, CRM Systems, Workflow Automation, Agile, Product Analytics, Stakeholder Communication | **Finance:** Financial Modeling, Market Forecasting

EXPERIENCE

- Dawn Adam & Co** — *Founder* Jun 2025 – Sep 2026 · Delhi, India (Hybrid)
- Founded and lead Dawn Adam & Co, a fintech startup, building the venture from the ground up and overseeing product direction, branding, and operations.
 - Grew and managed a team of 100+ employees while scaling the company.
- Eve Consultancy** — *Business Development Manager Intern* Jul 2026 – Aug 2026 · Remote
- Consulted directly with client businesses on financial and company-related concerns as a business development point of contact.
 - Built financial models for client companies to support their strategic and operational decisions.
- Polluxa** — *AI Product & Business Analyst Intern (Team Lead)* Jun 2026 – Aug 2026 · Remote (Part-time)
- Owned analytics and CRM workflows across a 3-product portfolio, turning KPI insights into recommendations presented directly to the founder and product stakeholders.
 - Automated data collection, preprocessing, and reporting pipelines in Python and SQL, cutting manual reporting time ~60% across 5 weekly workflows.
 - Delivered Power BI dashboards translating 10,000+ row datasets into executive-ready business recommendations.
- Gravity Engineering Services** — *AI Product & Data Scientist Intern* Jun 2026 – Aug 2026 · Remote (Part-time)
- Performed backend development and database optimization for Gravity Engineering Services, strengthening performance and reliability across core systems.
 - Contributed to software development for Polluxa's core product, building backend functionality and engaging directly with clients and board members on project updates.
- Maruti Service Masters — JJ Impex Delhi Ltd. (Maruti Suzuki)** — *Business Analyst Intern* Dec 2025 – May 2026 · Delhi
- Delivered business and data analysis for a Maruti Suzuki subsidiary over 6 months, identifying process inefficiencies and proposing data-driven improvements to management.
 - Designed and maintained recurring KPI reports tracking key operational metrics, improving cross-departmental management visibility.
- IBM** — *Project Intern* Aug 2025 – Oct 2025 · Remote
- Built Clarity, an Electron-based live-captioning accessibility tool for deaf and hard-of-hearing users, independently capturing microphone and system audio into real-time, speaker-labeled captions with automatic OpenAI/Gemini transcription failover.
 - Shipped transcript-grounded Recap, Translate, Ask, and Define features across four LLM providers (OpenAI, Anthropic, Gemini, Nvidia), and hardened the app with context isolation, a strict CSP, and an explicit preload API allowlist.
- Xebia** — *Project Intern* Jun 2025 – Jul 2025 · Remote
- Built a hybrid retrieval-augmented generation pipeline from scratch, fusing dense retrieval (Chroma + sentence-transformers) with sparse BM25Plus search via Reciprocal Rank Fusion and a cross-encoder reranking stage.
 - Built an LLM-judged evaluation harness scoring claim-level faithfulness and answer relevance via forced tool-calling, backed by a 9-test dependency-light smoke suite running in CI with no API key or GPU required.
- Maruti Sales & Service Delhi — Maruti Suzuki India Ltd.** — *Data Analyst Intern* Jun 2024 – Nov 2024 · Delhi
- Queried and extracted data via SQL across multiple databases to strengthen business reporting for Maruti Suzuki's Delhi sales division.
 - Built Power BI dashboards and visual reports communicating operational trends to senior leadership across 3 business units.
 - Recognized by the General Manager for strong analytical performance — commendation documented in formal completion certificate.
- Laadli Foundation** — *Web Development & Research Intern* Jun 2024 – Jul 2024 · Delhi
- Shipped 3 live features for the Foundation's e-commerce and fundraiser platforms, contributing to both frontend and backend development.
 - Collected, organized, and validated 5,000+ field research records to support impact reports and project planning.

PROJECTS

- RL Derivative Hedging — Deep Hedging vs. Black-Scholes** — Python · JAX · Optax · Gymnasium [Github](#)
- Trained a from-scratch PPO agent (JAX + Optax) to hedge options against two Black-Scholes baselines across 3 volatility regimes (GBM, Heston), evaluated on 3,000 paired paths per regime at two cost levels.
 - Validated against the closed-form Boyle-Emanuel convergence rate first; found the agent cuts turnover 30–45% vs. both baselines in every regime, beating the static hedge outright at higher trading costs.
- Neural Volatility Surface Forecaster — IV Surface Calibration & Forecasting** — Python · PyTorch · NumPy [Github](#)
- Built a pipeline from a raw options chain to an arbitrage-checked IV surface (vega-weighted, multi-start SVI) to a Heston model calibrated via a 200K-parameter neural surrogate trained on 20,000 samples.

- ▶ Achieved 0.54 vol-point surrogate RMSE with sub-second calibration to new surfaces; an LSTM forecaster beat a persistence baseline by 4.6% RMSE at a 5-day horizon.

S&P 500 Direction Prediction — Production ML Pipeline — Python · Pandas · Scikit-learn · XGBoost · LightGBM · Streamlit

[Github](#)

- ▶ Built a leakage-safe ML pipeline predicting next-day S&P 500 direction on 90 years of data (1927–2019); engineered 63 features and benchmarked 5 classifiers via 6-fold walk-forward validation.
- ▶ Caught and fixed a bug in the project's own look-ahead-bias checker via a 29-test pytest suite; reported honest results (52.8% accuracy vs. a 52.9% naive baseline).

Sightline — Grounded AI Analytics Copilot — Python · FastAPI · PostgreSQL · Next.js/TypeScript · Docker · Anthropic API

[Github](#)

- ▶ Architected a 5-step grounded AI-query pipeline (FastAPI, PostgreSQL, Claude API) that verifies every number against real query results before it ships, with column-level RBAC enforced at the schema layer.
- ▶ Identified and patched a real permission-bypass vulnerability in the access check, and shipped with a 53-test CI suite covering unit, integration, security, and LLM-eval coverage.

EDUCATION

University of Petroleum and Energy Studies

Aug 2023 – Jun 2027 (Expected)

B.Tech CSE, Data Science | Minor: Finance | Coursework: ML, DBMS, Statistics, Financial Modeling

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LEADERSHIP & ACHIEVEMENTS

- ▶ Founded and led Dawn Adam & Co, a fintech startup, growing and managing a team of over 100 employees.
- ▶ Head of Information Technology, Amrita Vidyalayam — led the school's IT operations in this part-time leadership role.
- ▶ Operations Team Lead, UPES MUN — led operations for the university's Model United Nations society.
- ▶ Team Lead for Polluxa's AI Product & Business Analyst internship, directing a 3-product analytics and CRM initiative.
- ▶ National-level Model United Nations (MUN) delegate during school.